Kyuseong Choi

Cornell Tech kyuseongchoi5.github.io kc728@cornell.edu

Education

Ph.D., Statistics and Data Science, Cornell Tech, New York *2021—*

Advisors: *Prof. Raaz Dwivedi & Prof. Kengo Kato*

M.S. Biostatistics, University of Michigan, Ann Arbor *2021*

Advisors: *Prof. Jeremy Taylor & Prof. Peisong Han*

B.B.A. Business Administration, Korea University, Seoul *2019*

Military service (Korean Air Force) 2014 — 2016

Research Interests

Benchmark for site reliability engineering, Post-training large language models for non-verifiable tasks, Foundational models for matrix completion, Distributional / personalized causal inference, High-dimensional CLTs, Kernel methods

Experience

Applied Scientist intern, Amazon P2 Science *2025.05-08*

Research: Kernel uniform inference for conditional average treatment effect under endogeneity

Work in progress

R1. Albert Gong, **Kyuseong Choi**, Abhineet Agarwal, Raaz Dwivedi, "ORCA-Bench: Are Code Agents ready for Oncall?". *2026*

R2. **Kyuseong Choi**, Kengo Kato, "Gaussian approximation in Reproducing Kernel Hilbert spaces, applications to Kernel Ridge Regression". *2026*

Preprints

P1. **Kyuseong Choi**, Dwaipayan Saha, Woojeong Kim, Anish Agarwal, Raaz Dwivedi, "GOPO: Policy Optimization using Ranked Rewards", *arxiv.* *2026*

P2. **Kyuseong Choi**, Gabriela Ciolek, "Wild regenerative block bootstrap for Harris recurrent Markov chains", *arxiv.* *2025*

P3. Jacob Feitelberg, Dwaipayan Saha, **Kyuseong Choi**, Zaid Ahmad, Anish Agarwal, Raaz Dwivedi, "TabImpute: Accurate and Fast Zero-Shot Missing-Data Imputation with a Pre-Trained Transformer", *arxiv.* *2025*

P4. **Kyuseong Choi**, Jacob Feitelberg, Anish Agarwal, Raaz Dwivedi, "Learning counterfactual distributions via kernel methods", *arxiv, R&R in AoS.* *2024*

P5. Jacob Feitelberg, **Kyuseong Choi**, Anish Agarwal, Raaz Dwivedi, "Distributional matrix completion via nearest neighbors in the Wasserstein space", *arxiv, under review at JASA.* *2024*

Workshops

W1. Caleb Chin, Aashish Khubchandani, Harshvardhan Maskara, **Kyuseong Choi**, Jacob Feitelberg, Albert Gong, Manit Paul, Tathagata Sadhukhan, Anish Agarwal, Raaz Dwivedi, "N^2: A Unified Python Package and Test Bench for Nearest Neighbor-Based Matrix Completion", *ICML CODEML.* *2025*

Publications

J1. Albert Gong, **Kyuseong Choi**, Raaz Dwivedi, "Supervised kernel thinning", *Neurips.* *2024*

J2. **Kyuseong Choi**, Jeremy Taylor, Peisong Han, "Robust data integration from multiple external sources for generalized linear models with binary outcomes", *Biometrics.* *2024*

J3. Jeremy Taylor, **Kyuseong Choi**, Peisong Han, "Data integration: exploiting ratios of parameter estimates from a reduced external model", *Biometrika.* *2023*

Contributed Presentations

Uniform inference for stationary distribution of diffusion process: wild regenerative block bootstrap for Harris recurrent Markov chains

- 2025 JSM, Nashville *Aug 2025*
- 2025 INFORMS, Atlanta *Oct 2025*

Test bench and unified python package for Nearest Neighbor-based matrix completion

- 2025 ICML CODEML Workshop, Seattle *July 2025*

Supervised kernel thinning

- 2024 Neurips, Vancouver *Dec 2024*

Learning counterfactual distributions via kernel methods

- 2025 ICCOPT, LA *July 2025*
- 2024 INFORMS, Seattle *Oct 2024*
- 2025 SCI, ACIC, Detroit *May 2025*

Gaussian approximation in Reproducing Kernel Hilbert spaces

- 2024 Workshop on Translational Research on Data Heterogeneity, St. Louis *Apr 2024*

Robust data integration in generalized linear models

- 2023 JSM, Toronto *Aug 2023*

Teaching Experience

T1. TA: Statistical principles for data science (ORIE 5160), *Cornell Tech.* *2024*

T2. TA: Linear algebra, Operations research and financial engineering, Theoretical statistics (MATH 1710, ORIE 5740, STSCI 4090), *Cornell University.* *2021 - 2024*

T3. TA: Intro to Regression, Intro to Biostatistics (BIOS 502, 521), *University of Michigan, Ann Arbor.* *2019–2020*

Selected Achievements

- Korea Foundation for Advanced Studies (KFAS) Program Scholarship in Business Administration Studies *2017 — 2018*

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